

Examining the Actors and Functions of an Airline Operations Center

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The operation and safety of the National Airspace System (NAS) is reliant on a multitude of actors and systems, which includes the members of the Airline Operation Center (AOC), whose primary job is to ensure flight safety (Sheremeta & Weitzel, 2005). The FAA's NextGen will constitute many changes within the NAS, and the effects of these changes need to be examined in respect to the AOC. However, before these changes can be examined in detail, it is necessary to have a baseline understanding of the current operations and actors of an AOC. **In this study, AOC actors that are critical to the day-to-day operation of an AOC were identified and a functional allocation was performed.** Using this research as a baseline, further research should be conducted to examine how those AOC actors and their functions will change with the implementation of NextGen.

INTRODUCTION

The safety of the National Airspace System (NAS) is reliant on a multitude of actors and systems. Among those systems and actions are the members of the Airline Operation Center (AOC), whose primary job is to ensure flight safety (Sheremeta & Weitzel, 2005). While the dispatch position is required by Federal Aviation Administration (FAA) regulation, most US airlines operate an AOC to ensure safety and improve efficiency of operations. Often called the third leg of the NAS, the AOC assists both flight crews and air traffic controllers (ATC) and are critical to day-to-day operation of an airline (Feigh & Pritchett, 2005).

The Airline Operation Center

The typical AOC is comprised of fifty to one hundred operators in various roles. The AOC is operated around the clock to support the flight crew and air traffic controllers (Pujet & Feron, 1998). The AOC is not limited to just the dispatcher role or planning flight, but incorporates a wide breadth of responsibilities. For example, Delta Airline's AOC contains "meteorologists, representatives and immediate links to maintenance, representatives and immediate links to flight operations, immediate links to air traffic control, and, as importantly, hundreds or perhaps thousands of computers that provide current and predicted position depictions for all aircraft in the system as well as current and

predicted weather depictions" (Garland et al., 1999). AOC actors interact directly with the flight crew and ATC and are critical to airline operations during not only emergency situations, but also to day-to-day operations.

The most prominent and well-known member of the AOC is the airline dispatcher. The dispatcher strategically plans flights two to three hours before the flight departure, communicate critical information to flight crew, and assist in re-routing aircraft during situations requiring diversions, such as weather or emergencies (Garland et al., 1999). They perform all these functions and more while facing intense time pressures and external performance pressures (Sheremeta & Weitzel, 2005).

The AOC and NextGen

The FAA is conducting a transformation of the NAS. The Next Generation Air Transportation System (NextGen) aims to improve the convenience and dependability of air travel while increasing safety and reducing environmental impact. One of the primary goals of NextGen is to provide a technologically advanced and modernized NAS. In order to achieve these goals, NextGen proposes many changes, which may affect the roles and responsibilities of the air traffic controller, flight crew, and airline dispatcher (FAA, 2010a). These changes will inevitably introduce new human error modes and human performance hazards into the NAS.

The changes brought about by NextGen are documented by the FAA in the NAS Enterprise Architecture (EA), which is comprised of various views that detail the activities, decision points, and schedule of NextGen operation improvements (Krois et al., 2010). The EA outlines these plans in a variety roadmaps – one being the Human System Integration (HSI) roadmaps. The HSI roadmap connects human factors issues with other NextGen systems and roadmaps through operational improvements and decision points (Durso, et al., 2010). Additionally, the HSI roadmap approaches the EA from a people perspective and is the only roadmap to identify changes in the roles and responsibilities of actors within the NAS (Krois et al., 2010).

Included among the collection of actors in the 2011 HSI roadmap is the airline dispatcher and the AOC, whose roles will certainly change with the implementation of NextGen systems and procedures (FAA, 2011). For example, the HSI roadmap identifies “Interactive Flight Planning” as a work environment change in automation for the dispatcher, which is associated with the NAS EA operational improvement “Provide Full Flight Plan Constraint Evaluation with Feedback”. The NAS EA is also comprised of operational views (OV) that describe integrated NextGen operations in differing levels of detail. The OV-6c Operational View Event-Trace Descriptions explain the key dynamics of the NextGen operations in the midterm environment (2018). Several OV-6c scenarios also identified changes that will potential affect the airline dispatcher.

While many studies have investigated the effects of these changes in the air traffic controller position and the flight deck (Krois et al., 2010), few studies have examined the changes resulting for the implementation of NextGen in regards to the role of the airline dispatcher, much less the AOC. However, in order to understand how the AOC will change with NextGen, it is necessary to understand how the AOC operates before the changes.

Purpose

In order to better determine how the AOC's and its members' responsibilities and roles within the NAS will change with the implementation of NextGen, it is necessary to obtain a clear picture of the current AOC operation. The critical actors of the AOC should be defined. Additionally, their roles within the AOC, and their interactions with flight crews and air traffic controllers should be determined.

METHOD

In order to assess how the implementation of NextGen will affect the members of the AOC, it is necessary to determine a baseline of understanding in the various member of the AOC. First, the members of the AOC were identified and defined. **Only those members who are critical to the day-to-day operation of the AOC and those functions that are frequent and critical to day-to-day operation will be included in this analysis.** Following the identification of actors, a functional allocation was conducted to determine the responsibilities of the AOC critical actors (Wilson & Morrisroe, 2005). AOC subject matter experts were utilized in the process of functional allocation. Serving as the primary source of information, an AOC manual from Airline X (2002) was utilized in defining actors, and the various AOC members were identified in current scientific literature where applicable.

RESULTS

First, the members of the AOC, who are critical in the daily operation, were identified and can be found in Table 1. In addition to the traditional name, alternate names for the various actors were listed. For each actor, his or her role within the AOC was summarized, and the requirement to possess a dispatcher license and currency was also noted. As can be seen in Table 1, the AOC extends beyond the traditional dispatcher role to incorporate ATC coordinators, crew schedulers, and maintenance managers.

Table 1: Select AOC Actors

AOC Positions [Actor]	Alternate Names (if applicable)	Role Description	Requires Dispatcher License/Currency	Reference
Aircraft Dispatcher	Flight Superintendent, Airspace Manager	• Shares joint responsibility for the operational safety of flight including ◦ Planning flight ◦ Monitoring flight in-progress ◦ Assisting flight crews ◦ Disseminating information to crew ◦ Communicating with crews during emergencies or off-nominal events	Yes/Yes	Ball et al., 1998 Garland et al., 1999 Idris et al., 2006 Moertl et al, 2007 Sheppard, 2000 Sheremeta & Weitzel, 2005 Smith et al., 1996 Smith et al., 2007 Pujet & Feron, 1998
Operations Manager	Ops Manager, Ops Supervisor, Dispatch Manager, System Ops Controllers	• Responsible for supervision of personnel in the AOC. • Conduct the “day of” operational plan of the duty director and oversees the system-wide operation.	Yes/No	Moertl et al, 2007 Sheppard, 2000 Pujet & Feron, 1998
AOC Duty Director	Duty Director, Duty Manager	• Ensure the AOC team provides for the safe, efficient, and economical operation of all company aircraft. • Resolve problems associated with planning, scheduling, systems, and applications of air transport support. • Assist with unexpected delays and other issues impacting success. • Oversight responsibilities of AOC shift employees.	No/No	Feigh & Pritchett, 2005
Flight Follower	Aircraft Dispatcher Assistant	• Assist dispatcher in tracking flight in-progress. • Ensures the aircraft arrives at intended point of destination. • Monitors flight progress in diversion scenarios.	No/No	
Aircraft Router		• Responsible for the safe, timely, and efficient routing of aircraft.	No/No	Pujet & Feron, 1998
Air Traffic Control Coordinator	ATC Coordinator	• Acts as point-of-contact between ATC and AOC. • Provides information and relief for flights that are adversely affected by ATC reroutes, delays, etc. • Assist dispatchers as necessary.	No/No	Idris et al., 2006 Sheppard, 2000 Smith et al., 2007 Pujet & Feron, 1998 Sheremeta & Weitzel, 2005
Manager Tactical ATC		• Responsible for efficient, constant, and scheduled operations of airline within the NAS. • Supports the ATC Coordinator.	No/No	
Crew Scheduler	Crew Scheduling Specialist, Scheduling Agent	• Works directly with the flight crews to resolve scheduling issues.	No/No	Sheppard, 2000 Sheremeta & Weitzel, 2005 Pujet & Feron, 1998
Crew Scheduling Duty Manager		• Responsible for shift management of the crew scheduling department.	No/No	
Maintenance Controller	Maintenance dispatch	• Acts as liaison between line maintenance and AOC by providing updated on maintenance issues. • Assist in resolving in-flight maintenance issues.	No/No	Sheremeta & Weitzel, 2005
Manager Maintenance Control		• Responsible for supervision of maintenance controllers. • Makes final decision on aircraft maintenance status.	No/No	

	Actors										
	Aircraft Dispatcher	Operation Manager	AOC Duty Director	Flight Follower	Aircraft Router	ATC Coord.	Manager Tactical ATC	Crew Scheduler	Crew Scheduling Manager	Maint. Controller	Manager Maint. Control
Plan Flight	X				•	•		•		•	
Monitor/Track Flight	X	•	•	X		•					
Communicate with Flight Crew	X	•						•	•	X	•
Communicate with ATC	•	•	•			X	•				
Plan aviation maintenance	•	X	•		•					X	•
Assist during diversions	X	X	X	X	•	•		•	•	•	•
Schedule flight crew	•	•	•					X	X		

Note: X-indicates actor plays major role in function. •-indicates actor plays minor role in function.

Figure 1: AOC Functional Allocation

The AOC actors were then compared to traditional AOC functions and the results can be seen in Figure 1. The degree to which each actor plays a role in each function is also indicated.

DISCUSSION

General Findings

As can be seen in Table 1, the AOC is not limited to just a dispatcher or flight planner, but incorporates an assortment of actors in the day-to-day critical operations. While some of the other AOC actors are present in literature, it is clear that the research has been focused on the dispatcher role and rarely incorporates the full spectrum of members of an AOC. For example, the flight follower assists the dispatcher by monitoring flights and alerting the dispatcher to problems with flights. While not all operations incorporate this role, most large AOCs utilize this position to transfer some of the responsibilities of the dispatcher to the flight follower. As such, the flight follower is not nearly as prevalent in the scientific literature as the dispatch role, but the safety implications of the flight follower role are evident and substantial.

While the dispatcher has the ability to perform many of the functions of the AOC, the main responsibility of the function does not lie with the

dispatcher, but rather is allocated to the other members of the AOC. For example, the ATC coordinator's primary obligation is to communicate with ATC and act as the singular point-of-contact with ATC. While the dispatcher may communicate with ATC in emergency situations, the dispatcher does not interact directly with ATC; rather information provided by the dispatcher and other members is relayed to ATC via the ATC coordinator.

NextGen Considerations

As previously mentioned, with the introduction of NextGen, the role and responsibilities of the controller, pilot, and AOC will be affected (Krois et al., 2010). Fortunately, some studies have investigated the affects of particular aspects of NextGen or a singular NextGen system. For example, distributed problem-solving systems have been investigated and the potential ramifications and inefficiencies related to the dispatcher were identified (Smith, Spencer, & Billings, 2007). However, the NextGen implications have yet to be examined at a higher systems level. In relation to the airline dispatcher, the HSI roadmap identified three work environment changes (4D departure management, interactive flight planning, and

improved conflict with resolution advisories added) and one mixed workforce/work environment change (full collaborative decision power). Therefore, it is suggested that efforts should be taken to investigate how these programs and other programs will affect not only the role of the dispatcher, but also the role of the AOC, as a whole and the individual members.

CONCLUSION

NextGen will constitute many changes within the NAS. The effects of these changes are already being examined in respects to the controller and the flight crew, but little effort has been made to examine the effect to the dispatcher, much less to the AOC. The purpose of this study was to achieve a baseline understanding of the current practice of an AOC. Actors that were critical to the day-to-day operation of an AOC were identified and a functional allocation was performed. Further research examining how those actors and their functions will change with the implementation of NextGen is necessary to the safety of the NAS.

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